

**ITS203 Object-Oriented Design and Programming**

# **Recipe & Pantry Manager**

## **Milestone 1 – Project Idea Proposal**

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## **1. Problem Description**

Many home cooks, especially students and young professionals living alone for the first time, often buy goods without verifying what they already have in their cupboard, fridge or freezer. This leads to double buying, unused products that expire and the ongoing agony of dinner planning: looking into a fridge full of “something” but not understanding what can actually be made with it.

Households account for approximately 60% of the world’s food waste, throwing away an estimated 631 million tons of edible food annually (United Nations Environment Program, 2024), and a four-person household in the United States wastes nearly \$3,000 worth of groceries each year, mostly food purchased but never used before it went bad (United States Environmental Protection Agency, 2025). One of the big drivers is just forgetting what is already at home. This is the case for anyone who cooks regularly, but especially students and early career professionals cooking on a budget with limited storage and irregular shopping habits – users who would benefit from an application that keeps an up-to-date record of what they have and helps them decide what to cook without having to manually take stock.

## **2. Proposed Idea**

The Recipe & Pantry Manager is a Windows Forms desktop program which allows a user to keep two connected collections, a personal recipe book and a live inventory of their pantry. Essentially, it has a matching engine that automatically compares the ingredients needed for each saved recipe against what's currently logged in the pantry, displaying at a glance which recipes can be cooked now, which are just missing a few ingredients, and which pantry items are nearing their expiry date and should be used soon turning "what can I make for dinner?" into a question the application answers directly.

## **3. Inspiration**

The idea comes from a behavior that every home cook experiences: You go to the cupboard for an ingredient for a meal and find you already have something almost exactly like it in the back of the shelf, or find vegetables that have gone bad because

you didn't have a plan for what to do with them. A pantry-and-recipe application is also a good candidate for the OOP concepts covered in this unit: Recipe, Ingredient, and Pantry Item are all natural, real-world classes, and the relationship between "ingredients needed" and "ingredients owned" presents a real design problem to solve, rather than an artificial one the main reason this project was chosen over a more generic option.

## **4. Functional Requirements**

The program will have following core features:

1. Users can upload, amend and remove recipes. Each recipe has a name, category and a list of essential ingredients with their proportions and units.
2. The user can add, update and delete pantry items, entering the item name, quantity, unit and (optionally) an expiry date.
3. The app will tell you if you can make each bookmarked dish with the current pantry stock, and if not, what precise ingredients you are lacking.
4. The user can search for recipes by name, category or a single ingredient and filter the search results.
5. The app highlights pantry goods that are expired or expiring within a preset number of days (e.g. 3 days) so the user can prioritize using them.
6. When a user marks a recipe as "cooked" the application automatically reduces the used quantity from the relevant pantry goods.

## **5. Non-Functional Requirements**

1. The application should be able to filter, search and recalculate "can I cook this?" answers without significant delay, for a pantry and recipe book of up to a few hundred items each.

2. Usability: The interface should be simple enough to allow a first time user to add a pantry item or recipe without instructions, with clear labels and sensible default values.
3. Reliability: Recipe and pantry data should be persisted properly between sessions. The program should “fail gracefully” (rather than crashing) on unexpected input (e.g. a missing quantity).

## 6. Data Storage Plan

The application will use file based storage instead of a database. Two JSON files will be stored in the application's data folder that will map directly on the object model that will guide the class design:

- `recipes.json` - a serialized `List<Recipe>` where each `Recipe` has an `Id`, `Name`, `Category` and a `List<RecipeIngredient>` (each with a `Name`, `Quantity` and `Unit`).
- `pantry.json` - a serialized `List<PantryItem>` with each `PantryItem` having `Id`, `Name`, `Quantity`, `Unit` and optional `ExpiryDate`.

The serialization will be done with .NET's built-in `System.Text.Json` package (Microsoft, n.d.) reducing the requirement for an external database. The reasons for choosing file-based storage rather than a database are threefold: the brief explicitly permits it as an equally valid choice at this stage; it avoids the setup and schema-migration overhead that comes with an engine such as SQLite, leaving more time for the OOP design and functionality actually being assessed; and, as the application is single-user and single-device, JSON files are easy to inspect and debug during development. Then later going to SQLite would be a natural extension, as the object model already separates data access behind a simple save/load interface – keeping the scope realistic within the remaining weeks.

## References

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